

Best Model Selection & Reasoning

Three machine learning models were evaluated: Linear Regression, XGBoost Regressor, Tuned XGBoost, and Random Forest using TimeSeriesSplit cross-validation.

Model	MAE	RMSE	MAPE	R ²	WMAE
Linear Regression	2386.92	6381.25	1.122×10^{15}	0.9135	2386.92
XGBoost	1654.18	4471.07	5.281×10^{14}	0.9577	1654.18
Tuned XGBoost	1752.10	4660.68	5.427×10^{14}	0.9539	1752.10
Random Forest	1717.35	5004.20	4.116×10^{14}	0.9467	1717.35

Best Model Selection

XGBoost Regressor was selected as the best model because it achieved the lowest MAE, RMSE and WMAE together with the highest R² score, indicating the best overall forecasting performance.

Trade-off Analysis

- Linear Regression: Highly interpretable but lower accuracy.
- XGBoost: Best accuracy, good scalability, moderate deployment complexity.
- Tuned XGBoost: Slightly slower due to GridSearchCV without improving accuracy.
- Random Forest: Robust and stable but slightly less accurate than XGBoost.

Business Context

XGBoost supports accurate inventory planning, reduces stock-outs and overstocking, scales well to large retail datasets, and provides fast predictions suitable for operational retail forecasting.

Limitations & Future Improvements

The model does not include external variables such as weather or competitor pricing. Future work could incorporate LightGBM, Prophet, LSTM, Optuna-based tuning, and automated retraining pipelines.